A Geometric Criterion for Sway and Rocking in Inclined Pile Groups

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Abstract

Operational modal analysis of a pile-supported foundation returns natural frequencies and mode shapes, but not the mechanism that each mode represents. For an elevated cap on inclined piles the question at the lowest frequency is whether the cap sways or rocks, which usually requires the soil stiffness, the least certain input. This paper shows that the geometry settles much of it. The six-degree-of-freedom stiffness of a rigid cap shows that the axial pile springs, which carry the soil stiffness, leave one motion almost unresisted: rotation about the point where the pile axes meet, here 42 to 45 m above the pile heads. The lowest lateral mode therefore rotates about a centre at or above that focal point and is resisted mainly by pile bending, and its vertical component at the cap is bounded by geometry alone. Rocking about a centre near the pile heads lies five to sixteen times higher, and torsion falls close to the lowest lateral pair. The prediction is tested on an offshore wind turbine foundation measured before and after tower installation. Of the identified poles, only one band, 1.05 to 1.33 Hz, meets three acceptance requirements derived from the definition of a mode. In the setup with a documented vertical channel, the vertical-to-horizontal amplitude ratio of that band is 0.08 to 0.16. Focal-point sway predicts 0.03 to 0.08 and rocking about 1. One member of the band has a torsional pattern. The measured frequency constrains the depth to fixity of the piles rather than their axial stiffness.

Keywords: inclined pile group; pile-group foundation; operational modal analysis; sway and rocking; offshore wind turbine; soil–structure interaction

1 Introduction

A modal identification algorithm fits a linear state-space model to measured covariances and returns its eigenvalues. Nothing in that operation distinguishes a mode of the structure from mains interference or from a resonance of the instrument mounting. All three appear as poles, all three can be made to stabilise cleanly across model orders, and all three can carry excellent quality indices. Stabilisation measures the internal consistency of a fit; it provides no evidence about what was fitted.

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Acceptance must therefore be imported from outside the fit. Two sources are available before any measurement is taken. The first is the definition of a mode: a pair of eigenvalue and mode shape satisfying the homogeneous equation of motion, in which the mass, damping and stiffness belong to the structure and not to the instrumentation. The second is the structure of the estimator itself, which can only resolve what its covariance window spans. Both yield requirements that can be applied to raw records without a validated model, a documented sensor layout or environmental data, which matters in construction-stage measurement, where none of the three is normally available.

A parallel difficulty arises on the modelling side. The lowest mode of a pile-group foundation may be a sway of the cap or a rocking of the cap. The question is usually settled by constructing a model, and such a model requires the soil stiffness. That is the least certain quantity in the problem: it depends on strain level, on installation effects and on time, and for a foundation measured during construction it is rarely known to better than a factor of two. It is shown below that for an inclined pile group the shape of the lowest mode does not require it. The axial pile springs, through which most of the soil stiffness enters, are nearly concurrent: their axes meet close to one point above the cap. Rotation about that point stretches no pile, so the axial springs fix where the lowest mode rotates but not how fast.

This paper derives that prediction from the six-degree-of-freedom stiffness of a rigid cap and states it in a form the records can test: a bound on the vertical component of the lowest mode, set by geometry alone. It also sets out acceptance requirements that follow from the definition of a mode and can be evaluated on raw records. The two are brought together on a construction-stage measurement of an inclined pile group.

Section 2 sets both parts against the work on which they draw. Section 3 describes the foundation, the ground and the records. Section 4 derives the prediction from the geometry. Section 5 identifies the modes and decides which poles belong to the structure. Section 6 combines the two independent lines of evidence.

2 Related work

Four bodies of work bear on the problem: the stiffness of inclined pile groups, field identification of wind turbine foundations, the acceptance of identified poles, and the contamination of ambient records by harmonic and local responses.

2.1 Stiffness of inclined pile groups

The stiffness model of Section 4 is classical. Hrennikoff [1] analysed a rigid cap on batter piles by resolving the axial and transverse resistance of each pile onto the displacements of the cap, which is the route by which inclination turns axial stiffness into lateral stiffness. Saul [2] cast the analysis as a matrix of stiffness coefficients from which the natural frequencies of a pile foundation also follow, and Poulos and Davis [3] give the standard treatment of pile groups. Implicit in those matrices is a condition that Section 4 makes central: piles whose axes pass through one point offer no axial resistance to rotation about that point. Randolph and Wroth [4, 5] provide the closed-form axial stiffness of a single pile and the axial interaction factors used in Section 6.2. The benefit of inclination depends on the axial capacity it mobilises: in centrifuge tests McVay et al. [6] found battered groups to carry 20 to 50 % more lateral load than plumb groups in dense sand, but little more in loose sand, where the tension capacity of the piles was limited.

Dynamic interaction within pile groups was established by Kaynia and Kausel [7], who found group impedances to depend strongly on frequency, and reduced to practice through the interaction factors of Dobry and Gazetas [8] and of Gazetas and Makris [9, 10]. For inclined piles, Gerolymos et al. [11] showed that batter piles may be beneficial or detrimental according to superstructure slenderness and pile–cap connection, and Giannakou et al. [12] that the response

depends on the balance between overturning moment and shear. Padrón et al. [13] computed the complete vertical, horizontal, rocking and cross-coupled impedance functions of inclined piles and pile groups with a coupled boundary-element and finite-element formulation. The same line of work has since examined the effect of the angle of inclination on the dynamic properties of piled structures [14], extended the impedances to layered soils [15], and provided ready-to-use vertical and rocking impedances of pile groups for offshore wind turbine foundations [16], while Goit and Saitoh [17] measured the interaction factors and impedances of inclined piles experimentally. Carbonari et al. [18] examined soil–structure interaction effects in bridge piers founded on inclined pile groups, and Dezi et al. [19, 20] tested near-shore piles in the field, under lateral impact and for pile-to-pile interaction. Huang and Zhong [21] derived the coupled horizontal–rocking impedance of partially embedded pile groups, the configuration of an elevated cap, and showed its effect on the resonance of offshore wind turbines, for vertical piles. Arany et al. [22,23] derived closed-form estimates of offshore wind turbine frequencies that include foundation flexibility.

These studies predict an impedance, a response or a frequency, and each requires the stiffness of the soil to do so. The near-concurrency of the pile axes of a practical inclined group, and its consequence for the shape of the lowest mode of an elevated cap, have not to the authors' knowledge been used to predict a measurable property of that mode. The result of Section 4 is narrower than an impedance analysis. It does not give the frequency, which still depends on the soil, but it bounds the vertical component of the lowest mode from the setting-out drawing alone, and that bound can be checked against records.

2.2 Field identification of wind turbine foundations

Output-only identification of offshore wind turbines is established. Devriendt et al. [24] automated operational modal analysis to track the support-structure modes of a turbine in the Belgian North Sea, Damgaard et al. [25] identified cross-wind modal properties of offshore turbines by full-scale testing, and Dong et al. [26] combined vibration monitoring with operational modal analysis of an offshore turbine structure. Weijtjens et al. [27] used resonance frequencies to monitor a monopile foundation, and Norén-Cosgriff and Kaynia [28] found the first natural frequency to fall and damping to rise with load level. Tcherniak et al. [29] examined where the assumptions of the method break down on operating turbines, and van Vondelen et al. [30] reviewed damping identification for offshore turbines. Separating a structural change from environmental and operational variability requires long records with environmental covariates [31, 32], which construction-stage campaigns rarely provide.

Measured frequencies of monopile-supported turbines commonly exceed design predictions. Sastre Jurado et al. [33] found the parked eigenfrequencies of a North Sea wind farm to be systematically underestimated by the design models and identified foundation stiffness as a key source of the discrepancy, and Winkler et al. [34] benchmarked soil–structure interaction models against monopiles measured before the tower was installed. Over 47 locations, Nernheim et al. [35] found design frequencies between about 85 and 97 % of the measured values, a gap that stiffening the soil springs alone did not close.

Measurements during construction are fewer. Fernandes et al. [36] tracked resonance frequencies through every installation stage of 48 monopile-supported turbines, Seo et al. [37] tested a turbine on tripod suction buckets at four construction stages, and Louw and Kearsley [38] monitored the strains of an onshore piled raft through turbine installation and its first year of operation. Pile groups under an elevated cap, the foundation of the Donghai Bridge wind farm in China and of the nearshore farms of the Mekong Delta [39, 40], have been studied mainly for bearing capacity [41]. Their dynamics have been examined in centrifuge tests of a turbine on a group-pile foundation [42] and by transfer-path analysis of the energy reaching an elevated pile cap [43]. Li et al. [44] identified the modes of an operating 4 MW turbine on such a foundation, eight piles inclined at 6:1 under a 15 m cap, from accelerometers distributed along the tower, and found the first mode at 0.36 Hz; Liu et al. [45] studied a cap–pile group of the Donghai

Bridge farm by experimental modal analysis. In bridges, a similar configuration, a pier on piles standing in water, has been tested modally by Deng et al. [46]. In Vietnam, Huynh [47] modelled a single pile foundation as a rigid body to study the loads from blade breakage. These studies identify the modes of the turbine as a whole, measured along the tower or at a single level. None that was found instruments the cap of an inclined group so that its pattern of motion, and not its frequency alone, can be compared with a mechanism predicted beforehand.

2.3 Acceptance of identified poles

Covariance-driven subspace identification [48, 49] and frequency domain decomposition [50] are the established output-only methods, and Reynders [51] reviews them together with the assumptions behind them. Stabilisation is the usual basis for accepting a pole, and the literature that defines it is explicit about what it tests. Peeters and De Roeck [52] describe it as discarding the spurious poles of an over-specified model by requiring agreement with the model one order lower, Döhler and Mevel [53] as separating physical from spurious modes by their constancy across orders, and Zhang et al. [54] judged its power to discriminate limited. Automated procedures add hard and soft validation criteria and cluster the stable poles [55–58]; modal phase collinearity [59] and the modal assurance criterion [60] test the consistency of each shape, and uncertainty bounds quantify the scatter of each estimate [53, 61]. Apart from limits on damping, these tests concern the consistency of the fit and the quality of each pole within one record. They do not test properties that a structural mode must have independently of the fit, such as extending over the instrumented structure or persisting when the sensors are moved.

Invariance across setups enters multi-setup identification as an assumption. Reference-based identification [49] rests on the invariance of an eigenvalue to the sensor configuration, and the merging schemes of Mevel et al. [62] and Döhler and Mevel [63] fit one state matrix to all setups because the same structure is observed in each. Döhler et al. [64] noted that matching modes between setups after identification is fragile and that the assumption may fail in practice. Section 5.1 uses the same invariance as a test that a candidate pole must pass, not as a constraint imposed on the fit.

The number of block rows, which sets the covariance window of the estimator, is known to bias the estimates [65], and Priori et al. [66] proposed rules for its minimum value. Section 5.1 states the corresponding limit as a heuristic bound on the frequencies the estimator can resolve, and Table 6 shows that a window below that bound suppresses the admitted band. Subspace identification has been shown to be robust to white excitation whose covariance varies in time [67], but not to impulsive or coloured excitation, under which damping estimates show larger variance and bias [68]; damping is therefore not interpreted here.

2.4 Harmonic and local contamination

A deterministic harmonic can be identified as a structural pole [69], and several indicators separate the two. Brincker et al. [70] proposed an indicator based on the probability density of the response, which Jacobsen et al. [71] and Andersen et al. [72] applied as the kurtosis of band-pass filtered signals, about 1.5 for a sinusoid against 3 for Gaussian noise. Jacobsen [73] noted that a harmonic behaves as a very lightly damped mode, so that damping is itself a harmonic indicator and a valid range of damping in a stabilisation diagram removes harmonics together with non-physical poles. Modified identification algorithms [74], time–frequency decomposition [75] and projection of the periodic part out of the data [76] have also been used, and Motte et al. [77] reviewed the subject. These studies concern the harmonics of rotating machinery and rotors. The harmonics in the present records are electrical, introduced into the measuring chain at the mains frequency through ground loops [78, 79], but they present to the estimator in the same way. The damping requirement of Section 5.1 follows Jacobsen [73], and its bound is set there against tabulated structural damping.

A resonance need not belong to the structure the sensors were placed to characterise. The mounting of an accelerometer alters the response it records [79,80], and local modes at frequencies close to global ones complicate identification, as Abdelnour and Zabel [81] showed for a space truss, separating the two by a modal strain energy indicator. The effective channel count of Section 5.1 is the participation ratio introduced by Bell and Dean [82] to measure the spatial extent of vibrational modes, and localisation of vibration by structural irregularity was established by Hodges [83] and by Pierre and Dowell [84]. No use of the participation ratio as an acceptance criterion in operational modal analysis was found.

Each of the three requirements of Section 5.1 therefore has an antecedent: the damping bound in the harmonic indicators of Jacobsen, invariance across setups in multi-setup identification, and the effective channel count in the participation ratio of localisation physics. What was not found is their joint use as acceptance tests on construction-stage records, or a mode-shape prediction derived independently of the records against which the accepted mode can be checked.

3 The offshore wind turbine foundation and the field measurements

The geometry and the soil profile are read from the design drawings and from the borehole sunk at this turbine. The measurement inventory and the excitation characteristics come from the raw records of two ambient vibration campaigns, one recorded before the tower was erected and one after.

3.1 Foundation geometry and ground conditions

The foundation carries turbine T18 of the Dong Hai 1 wind farm, Tra Vinh, Vietnam. It consists of a circular reinforced concrete cap, 20.5 m in diameter and 4 m thick, supported on 34 prestressed high-strength concrete piles of 1.0 m outside diameter and 46 m length, arranged in three concentric rings. The cap soffit is at elevation +5.435 and the seabed at −6.900, so each pile stands free in water over 12.34 m before entering the ground; the toe is at approximately −40.5.

Two quantities govern the central result of Section 4: the radius of each ring and the inclination of the piles in it. Together they fix the height $r \cot \alpha$ at which the axes of each ring meet. Nothing else about the pile enters that height: not its section, not its material, not the soil around it. The section, the material and the soil enter the frequencies instead.

The arrangement was read from the pile setting-out sheet and is reproduced in Figure 1. It was then checked against the dimension string printed along the top of the same sheet. Measured inward from the cap edge, its successive offsets are 1250, $543 + 1563 + 894$, $804 + 696 + 1500$ and $1437 + 1563$ mm, which add up to the three ring radii of 9.0, 6.0 and 3.0 m and to the cap centre. The intermediate grid lines decode in the same way, matching $9.0 \cos 20°$, $9.0 \cos 40°$, $6.0 \cos 30°$ and so on. The layout in Table 1 is therefore a decoding of the dimensions written on the drawing rather than a measurement from a scanned image.

Two lengths determine the dynamic behaviour of the ground beneath the foundation. The pile is unsupported over 12.34 m of water, and the first 13.4 m below the seabed is very soft fat clay with SPT N never exceeding 2, in which it receives almost no lateral restraint. Together these place the whole flexible part of each pile in its upper third. The competent strata below — very stiff lean clay from 16.0 m, medium dense clayey sand to 30.5 m and very stiff fat clay beneath that — act only over the lower half.

This accounts for the fundamental frequency of this foundation falling near 1 Hz rather than in the range expected of a cap bearing on firm ground. The profile and the SPT record are reproduced in Figure 2.

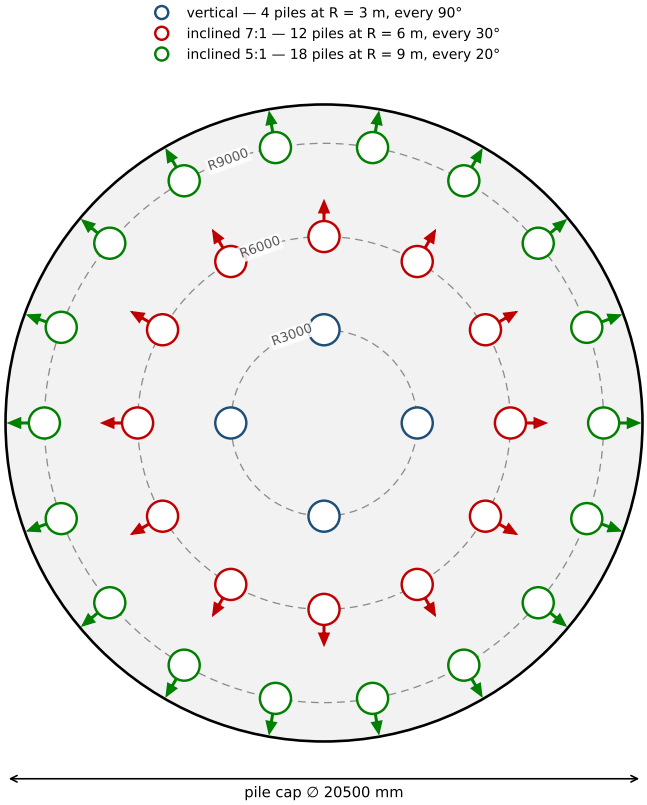


Figure 1: Pile layout of the T18 foundation, redrawn from the setting-out sheet *D1000 PHC pile coordinate*: 34 PHC D1000 piles on three concentric rings of radius 3.0, 6.0 and 9.0 m, carrying 4, 12 and 18 piles respectively. Arrows give the direction of inclination; the inner ring is vertical, the middle ring is inclined 7:1 and the outer ring 5:1. The cap is 20.5 m in diameter

Table 1: Foundation geometry and pile section properties, all read from the design drawings. The section properties enter the frequencies of Section 4.2 and the fixity analysis of Section 6.2; they do not enter the focal heights.

Cap	20.5 m diameter, 4 m thick; soffit +5.435, top +9.435
Inner ring	$R = 3.0$ m, 4 piles at 90°, vertical
Middle ring	$R = 6.0$ m, 12 piles at 30°, inclined 7:1 (8.1° from vertical)
Outer ring	$R = 9.0$ m, 18 piles at 20°, inclined 5:1 (11.3° from vertical)
Pile	prestressed high-strength concrete, 1.0 m outside diameter, 46 m long; toe at ≈ -40.5
Free length in water	12.34 m (cap soffit +5.435 to seabed −6.900)
Wall thickness	130 mm $\rightarrow A = 0.3553$ m^2, $I = 0.03437$ m^4
Elastic modulus	40 GPa $\rightarrow EA = 1.42 \times 10^{10}$ N, $EI = 1.37 \times 10^{9}$ N m^2

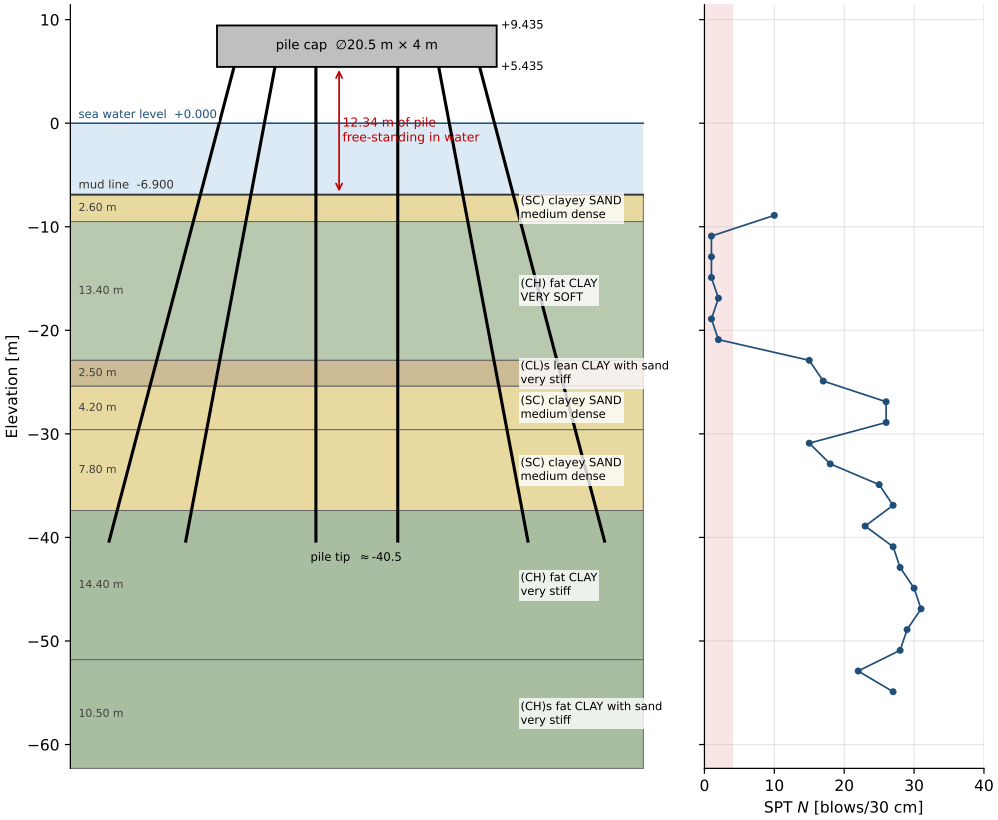


Figure 2: Foundation section and the soil profile at borehole DH9, with the SPT record alongside. Elevations are in metres. Each pile stands free in water over 12.34 m between the cap soffit at +5.435 and the seabed at −6.900. The fat clay immediately below the seabed is very soft, with $N \leq 2$ over 13.4 m

3.2 Instrumentation and measurement campaigns

Two ambient vibration campaigns were recorded on the top surface of the cap: campaign 1 before the tower was erected and campaign 2 after. Each comprises several setups differing in sensor position and measurement direction, seven in total, with three to six accelerometers recorded for between 9 and 26 minutes. Campaign 1 was recorded on 18 May 2024 and campaign 2 on 13 July 2024, each within about one and a half hours. The records were acquired with a National Instruments system under LabVIEW and stored in TDMS format, in units of g, at $51\,200/31 = 1651.6$ Hz, the rate stated in the file metadata. The calibration factors recorded in the processing log differ by at most 3 % between channels. The accelerometer model, its mounting and the grounding of the measuring chain are not documented. The instrumented area is a square frame of 5×5 m with sensors at the four corners and at the centre and, in campaign 2, a sixth channel two metres above the centre on the tower itself. Figure 3 shows the arrangement. The position of the frame on the 20.5 m cap is not documented, and it is drawn concentric on the inference that the sixth channel, being mounted on the tower, lies close to the cap centre.

The excitation is not the process the identification method assumes, and the departure is larger in campaign 1 than in campaign 2. Figure 4 shows one campaign 2 record. A single event in it reaches 139 standard deviations of its channel and carries the kurtosis of a nine-minute record to 3883, and the running RMS shows that the event is confined to one channel. Of the signal energy, 78.6 % lies between 50 and 200 Hz and 0.7 % in the structural band from 0.2 to 5 Hz. Table 2 shows that campaign 1 is worse on two counts. In the lateral 2 setup the segment RMS varies by a factor of 112 within one record. In the lateral 1 setup one channel is 47 times

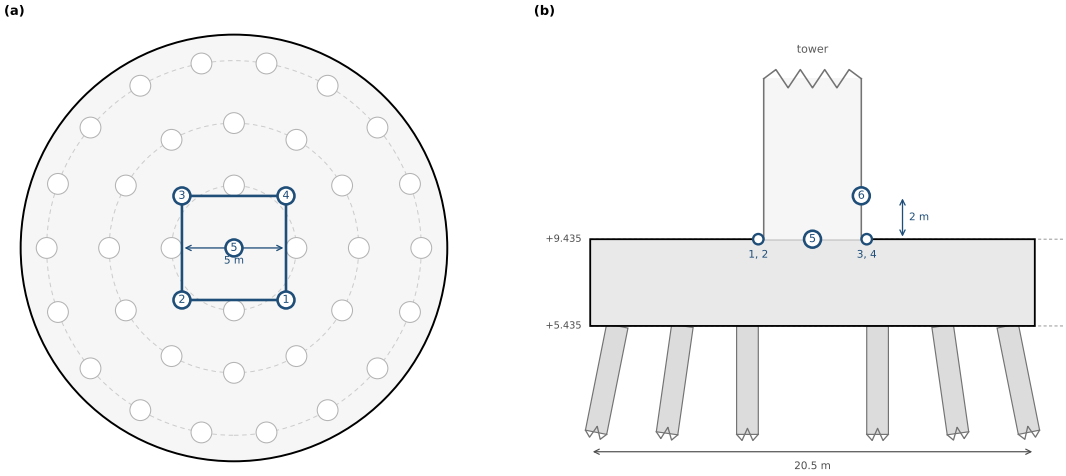


Figure 3: The instrumented frame. (a) Plan of the 5×5 m frame on the 20.5 m cap, with channels 1 to 4 at the corners and channel 5 at the centre. (b) Elevation, showing the sixth channel of campaign 2, which is mounted on the tower two metres above the cap surface and so cannot be placed in the plan. Only the six piles cut by a diametral section are drawn, and both they and the tower are shown broken off rather than at full length. The vertical setups measure Z on every channel. In the lateral setup of campaign 2 channels 1, 2, 3, 5 and 6 measure horizontally in one common direction and channel 4 vertically, as recorded in the processing log; the channel orientations of the two lateral setups of campaign 1 are not documented. The position of the frame on the cap is not documented, and it is drawn concentric on the inference that channel 6, being on the tower, lies close to the cap centre

Table 2: Measurement setups and raw-record statistics. *Ch.* is the number of channels and *Min.* the record length in minutes. *Peak*/σ is the largest absolute sample of any channel divided by that channel's standard deviation; *Kurt.* the largest channel kurtosis; $N_{>8\sigma}$ the largest number of samples of one channel beyond 8σ; *Drift* the ratio of largest to smallest channel RMS over ten equal segments; *spread* the ratio of the largest to the smallest channel standard deviation within a setup.

Camp.	Setup	Ch.	Samples	Min.	Peak/σ	Kurt.	$N_{>8\sigma}$	Drift	Spread
1	Vertical	5	2 535 936	25.6	133	562	2238	3.2	6.1
1	Lateral 1	5	1 523 873	15.4	25	54	3131	12.9	47.0
1	Lateral 2	5	1 385 189	14.0	50	1812	1063	**112**	9.2
1	Three-channel	3	1 545 336	15.6	10	4	9	1.5	33.7
2	Lateral	6	906 399	9.1	**139**	**3883**	724	4.7	11.5
2	Vertical d1	6	1 875 536	18.9	69	95	695	4.4	4.1
2	Vertical d2ke	6	2 260 219	22.8	67	88	**2564**	4.3	4.2

weaker than another in the same setup, so that setup carries fewer usable channels than its nominal count suggests. Only the three-channel setup of campaign 1 approaches stationary broadband behaviour, with a peak of 10σ and a kurtosis of 4.

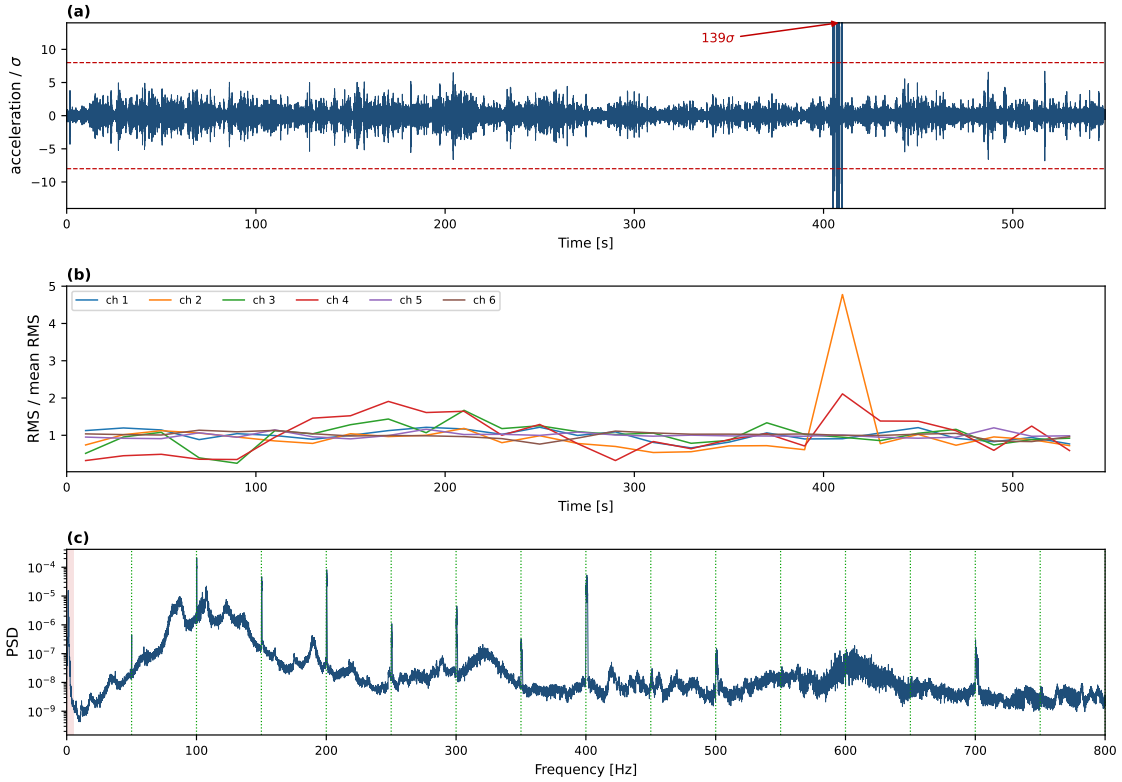


Figure 4: One record of campaign 2 in three views. (a) The full time history of the channel with the largest peak-to-standard-deviation ratio, in units of that channel's standard deviation and with the $\pm 8\sigma$ band marked. The vertical scale is clipped at $\pm 14\sigma$ so that the ambient level remains legible; the single event at 408 s reaches 139σ, carries the record kurtosis to 3883, and is marked by the arrow. (b) The running RMS of every channel over 20 s windows, normalised by its own mean; the event of panel (a) is confined to one channel. (c) The channel-averaged spectrum, with multiples of 50 Hz dotted; the band from 50 to 200 Hz carries 78.6 % of the energy and the shaded structural band from 0.2 to 5 Hz carries 0.7 %

Two consequences follow. Damping ratios are reported where the identification produces them, because the stabilisation criterion uses them, but they are not interpreted. Subspace damping estimates under impulsive, non-stationary excitation are biased by an unknown amount, and field studies of offshore support structures report damping that varies with load level and operating state [28]. Frequency estimates, in addition, carry more uncertainty than the scatter within a stabilisation cluster suggests, and for that reason Section 5.2 does not rest on frequency agreement between setups.

4 The lowest modes of an inclined pile group

The derivation that follows uses no measurement. It assembles the stiffness of a rigid cap on inclined piles in all six degrees of freedom, separates the part contributed by the axial pile springs, and shows what that part fixes regardless of the soil. Because the result is obtained before any record is examined, the comparison in Section 6.1 is a test and not a fit.

4.1 Idealisation and the stiffness of a rigid cap on inclined piles

The cap is treated as a rigid body. The idealisation is appropriate here because the cap is 4 m thick over a 20.5 m span while the piles are 1.0 m in diameter over 46 m: the cap is orders of magnitude stiffer in the relevant deformation than the elements supporting it. The cap has six degrees of freedom, collected in $\mathbf{q} = (\mathbf{U}, \mathbf{\Phi})$: a translation $\mathbf{U}$ and a rotation $\mathbf{\Phi}$ about the origin O, taken at the centre of the cap soffit, where the pile heads are. Pile i has its head at $\mathbf{p}_i$, at radius r_i and azimuth θ_i, and is inclined outward at α_i from vertical. Its axis has the unit vector $\mathbf{a}_i = (\sin\alpha_i \cos\theta_i,\ \sin\alpha_i \sin\theta_i,\ -\cos\alpha_i)$, pointing from the head to the toe. The head is rigidly connected to the cap, so it displaces by $\mathbf{U} + \mathbf{\Phi} \times \mathbf{p}_i$ and rotates by $\mathbf{\Phi}$.

Each pile is represented by an axial spring k_{ax} along $\mathbf{a}_i$ and by a beam of bending stiffness EI and torsional stiffness GJ, fixed at an effective fixity length L_{eff} measured along its axis. The beam gives the head the transverse stiffness $12EI/L_{\mathrm{eff}}^3$, the rotational stiffness $4EI/L_{\mathrm{eff}}$ and the coupling $6EI/L_{\mathrm{eff}}^2$ between them. Writing $\mathbf{k}_i$ for this 6×6 head stiffness in the axes of the pile and $\mathbf{\Lambda}_i$ for the rotation from global to pile axes,

$$\mathbf{K} = \sum_i \mathbf{B}_i^{\mathsf{T}}\, \mathbf{k}_i\, \mathbf{B}_i, \qquad \mathbf{B}_i = \begin{bmatrix} \mathbf{\Lambda}_i & \mathbf{0} \\ \mathbf{0} & \mathbf{\Lambda}_i \end{bmatrix} \begin{bmatrix} \mathbf{I} & -\mathbf{S}(\mathbf{p}_i) \\ \mathbf{0} & \mathbf{I} \end{bmatrix}, \tag{1}$$

in which $\mathbf{S}(\mathbf{p})$ is the skew matrix of the cross product. Nothing in (1) decouples translation from rotation; all cross terms are retained.

The mass matrix is that of the rigid cap, 3.30×10^6 kg with its centroid 2 m above O, together with a participating share of the piles and of the water they carry. That share is lumped at the pile heads. The central value is 13/35 of the mass of each pile over L_{eff} and of the added water over the free length, the equivalent mass of a beam fixed at one end and guided at the other; the bounds 0 and 1 are carried through every result.

4.2 What the axial springs fix

The axial springs contribute $k_{\mathrm{ax}}\, \mathbf{g}_i \mathbf{g}_i^{\mathsf{T}}$ to $\mathbf{K}$, with $\mathbf{g}_i = (\mathbf{a}_i,\ \mathbf{p}_i \times \mathbf{a}_i)$. The axial part of the stiffness is therefore $k_{\mathrm{ax}}\mathbf{G}$, in which $\mathbf{G}$ depends on the pile positions and inclinations alone. For translation u in x and rotation φ about y,

$$\begin{aligned} G_{uu} &= \sum_i \sin^2\alpha_i \cos^2\theta_i, \qquad G_{u\varphi} = \sum_i r_i \sin\alpha_i \cos\alpha_i \cos^2\theta_i, \\ G_{\varphi\varphi} &= \sum_i r_i^2 \cos^2\alpha_i \cos^2\theta_i. \end{aligned} \tag{2}$$

The axial extension of pile i under (u, φ) is $\cos\alpha_i \cos\theta_i\,(u \tan\alpha_i + r_i\varphi)$. For one ring of piles sharing a radius and an inclination, it vanishes for every pile at once when $u = -r\varphi\cot\alpha$. That is a rotation of the cap about the point on its axis at height

$$z_f = r \cot\alpha, \tag{3}$$

where the axes of the ring meet. Such a ring offers no axial resistance to rotation about its focal point, and its block of $\mathbf{G}$ is singular. Figure 5(a) shows the motion. A group of several inclined rings is singular only if their focal points coincide. In the T18 group they lie close together, because the steeper rake is given to the outer ring. For the T18 group the middle ring meets at 42.0 m above the pile heads and the outer ring at 45.0 m. The inner ring is vertical and contributes to $G_{\varphi\varphi}$ alone.

Table 3 gives the values. The coupling coefficient $G_{u\varphi}/\sqrt{G_{uu}G_{\varphi\varphi}}$ is 0.990, so translation and rotation cannot be treated separately. The two eigenvectors of the block are rotations about centres at 45.1 m and at -0.02 m. About the first centre the axial springs provide a rotational stiffness of $G_{\varphi\varphi} - G_{u\varphi}^2/G_{uu}$ per unit k_{ax}, 2 % of what they provide about the pile heads.

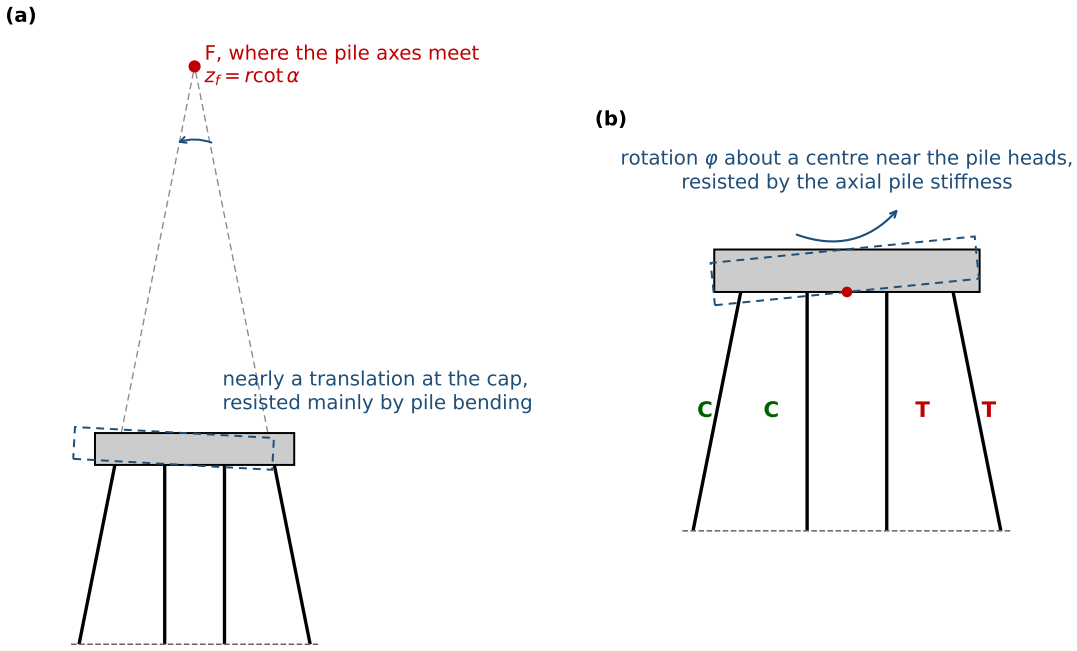


Figure 5: The two lateral mechanisms of a rigid cap on an inclined pile group. (a) Rotation about the focal point F at which the axes of the inclined piles meet, which stretches none of them; at the cap it is nearly a translation, and it is resisted mainly by pile bending. (b) Rocking about a centre near the pile heads, resisted by the axial pile stiffness acting on the lever arms. Dashed outlines give the displaced position; C marks a pile in compression and T a pile in tension. Only one ring of inclined piles is drawn

Three consequences follow, and each holds whatever the value of k_{ax}.

The lowest lateral mode rotates about a centre near or above the focal point. The axial springs cannot resist that motion, so its stiffness comes mainly from pile bending, and its centre of rotation z_c stays close to z_f. Bending adds lateral restraint at the pile heads and moves the centre further up, towards pure translation. At the top of the cap, at height h above the pile heads, a point at horizontal distance d from the rotation axis then moves vertically by $d/(z_c - h)$ of the horizontal motion. With $z_c \geq z_f$ this ratio is bounded by

$$\left.\frac{w}{u}\right|_{\mathrm{top}} \leq \frac{d}{z_f - h}, \tag{4}$$

a bound fixed by the setting-out drawing alone. For the T18 cap, $h = 4$ m, and a sensor at the corner of the instrumented frame, at d between 2.5 and 3.5 m, gives a bound between 0.06 and 0.09.

Rocking lies far above. The second eigenvector rotates about a centre near the pile heads and is resisted by the axial springs. At the cap top it moves the corners vertically by about as much as the cap moves horizontally. The vertical-to-horizontal ratio therefore separates the two mechanisms by an order of magnitude, and the bound (4) is the geometric criterion that this paper applies to the records.

Torsion lies close to the lowest lateral mode. Outward-inclined piles lie in radial planes, so a twist of the cap moves each head perpendicular to its axis and stretches none of them. Torsion is resisted by pile bending alone, as the lowest lateral mode mainly is, and the two cannot be far apart.

The frequencies themselves are not fixed by the geometry. Table 3 and Figure 6 give them over the full range of the uncertain inputs. The axial stiffness was varied from 1.5 to 6×10^8 N/m per pile, which brackets the estimates of Section 6.2. The fixity length was varied from 3 to 12

pile diameters below the seabed, and the pile mass share between its bounds. The lowest lateral pair falls between 0.62 and 1.34 Hz. It depends on L_{eff} through pile bending and hardly at all on k_{ax}: a fourfold increase in k_{ax} raises it by 5 %. Its centre of rotation lies between 48 and 105 m above the pile heads, never below the focal points, so the bound (4) holds throughout the sweep. Torsion lies between 1.09 and 1.21 times the lowest lateral pair. The rocking pair lies between 5.4 and 12.3 Hz, 5.0 to 16 times the lowest lateral pair, and it scales with k_{ax}. The vertical mode lies with it.

Table 3: The axial part of the stiffness of the T18 group, which depends on geometry alone, and the modes of the full model over the swept range of axial stiffness (1.5–6 × 10^8 N/m per pile), depth to fixity (3–12 pile diameters) and participating pile mass (0–1 of the pile over L_{eff} and the added water). The vertical-to-horizontal ratio is evaluated at the cap top for a point 2.5 or 3.5 m from the rotation axis

Axial springs only, per unit k_{ax}	
G_{uu}, $G_{u\varphi}$, $G_{\varphi\varphi}$, Eq. (2)	0.466 ; 20.6 m ; 930.6 m^2
Coupling coefficient $G_{u\varphi}/\sqrt{G_{uu}G_{\varphi\varphi}}$	0.990
Focal height $r\cot\alpha$, middle and outer ring	42.0 m ; 45.0 m
Centres of rotation of the two eigenvectors	45.1 m ; −0.02 m
Full model, axial springs with pile bending and torsion	
Lowest lateral pair, f_1	0.62–1.34 Hz
centre of rotation above the pile heads	48–105 m
vertical/horizontal at the cap top, $d = 2.5$; 3.5 m	0.025–0.057 ; 0.035–0.081
Torsion, f_{tor}/f_1	1.09–1.21
Rocking pair, f_{rock}	5.4–12.3 Hz
f_{rock}/f_1	5.0–16
centre of rotation above the pile heads	0.9–1.7 m
vertical/horizontal at the cap top, $d = 2.5$; 3.5 m	0.8–1.1 ; 1.1–1.5
Increase of f_1 for a fourfold increase of k_{ax}	5 %

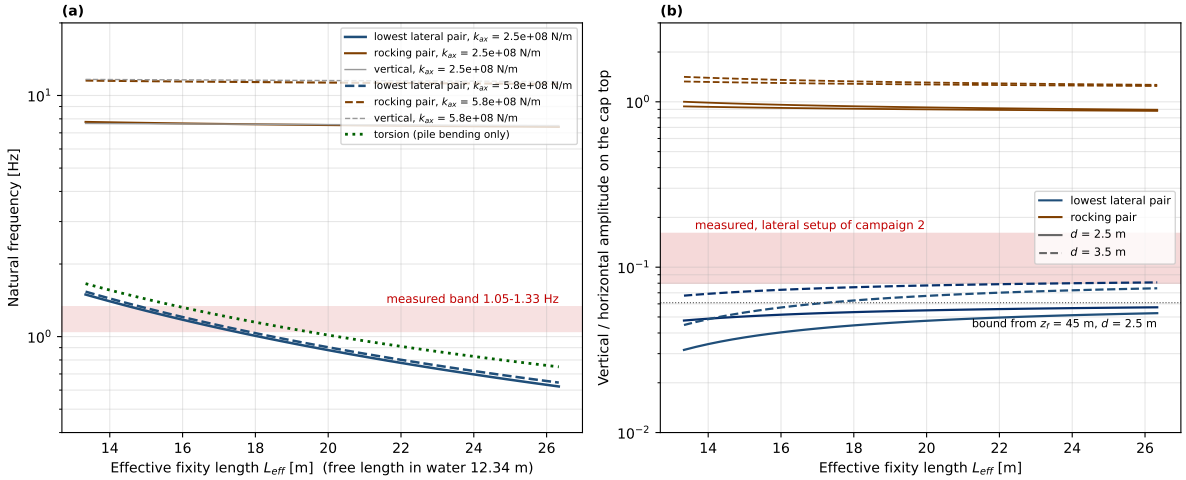


Figure 6: Modes of the rigid-cap model of the T18 group against the effective fixity length, for axial pile stiffness 2.5 and 5.8 × 10^8 N/m (solid and dashed). (a) Natural frequencies, with the measured band shaded. The lowest lateral pair and torsion are governed by pile bending; rocking and the vertical mode by the axial stiffness. (b) Vertical-to-horizontal amplitude ratio at the cap top, for points 2.5 and 3.5 m from the rotation axis. The dotted line is the bound (4) from the focal height at $d = 2.5$ m; the shaded strip is the range measured in Section 6.1

The result rests on the idealisation of Section 4.1, and the limits belong with it. It assumes a

rigid cap, small motion, one axial stiffness common to all piles and static rather than frequency-dependent pile stiffness. It also represents the restraint of the soil on each pile by fixity at a single depth. Full impedance analyses of inclined pile groups show that group stiffnesses depend on excitation frequency, on spacing and on the direction of inclination, and that horizontal–rocking cross-coupling can be large and can change sign with configuration [13]. The focal height does not depend on any of these, since it is a property of the pile axes. What can change is how close to it the lowest mode rotates, and the frequencies of the modes. The prediction that survives is therefore qualitative but testable: a near-translational lowest pair with a small vertical component bounded by (4), torsion close to it, and rocking well above.

5 Modal identification

Identification supplies candidate poles; it does not supply modes.

5.1 Identification procedure and acceptance criteria

Covariance-driven subspace identification assembles a block Toeplitz matrix of output covariances up to lag $2i$ and recovers the state and output matrices from its singular value decomposition [48, 49]. The estimator therefore observes the record through a time window $\tau = i\,\Delta t$, and this is a property of the construction rather than a processing choice. A mode of period T enters the covariances through their oscillation and decay across that window; when $\tau \ll T$ the covariance varies negligibly over the observed lags and carries nothing that distinguishes the mode from a constant offset. The block Toeplitz matrix reaches lags up to $2i$, so the span actually observed is 2τ, and the statement is a heuristic bound in the spirit of the rules of Priori et al. [66], not a sharp limit. It is nonetheless fixed, for a given configuration, before the data are consulted.

The processing chain follows from that requirement. Retaining the acquisition rate of 1651.6 Hz with $i = 150$ gives $\tau = 0.091$ s, which is 9 % of one period at 1 Hz and places any such mode outside the span of the estimator. The records are therefore decimated in two stages to 51.61 Hz with zero-phase anti-alias filtering, then high-pass filtered at 0.3 Hz with a fourth-order Butterworth. Identification uses $i = 120$ over model orders 2 to 80, which gives $\tau = 2.33$ s, or 2.3 periods at 1 Hz. Poles are accepted into a cluster when successive orders agree to within 1 % in frequency, 5 % in damping and 0.98 in modal assurance criterion, and a cluster is retained when it contains at least eight poles. A frequency domain decomposition of the same records [50] provides an independent check. Section 5.2 tests the resolution argument directly.

Three requirements are used as acceptance criteria. Each follows from the definition of a mode rather than from a convention of practice, each is stated as a property of the structure, and each can be evaluated on the raw records alone.

The first requirement is that a mode moves more than one measurement point. The mode shape is defined over the structure. If the observed shape has support on one channel and noise elsewhere, then the motion is confined to a region smaller than the sensor spacing, and the pole describes that region rather than the structure the instrumentation was placed to characterise. The measure used is the effective channel count $N_{\mathrm{eff}} = (\sum |\phi|^2)^2 / \sum |\phi|^4$, taken from the first singular vector of the cross-power spectral density matrix. It equals the number of channels when the shape is spread evenly over them, and unity when one channel carries the entire response. The rejection threshold is established by simulation. Synthetic records of 28 000 samples at the decimated rate were analysed with the same spectral resolution as the measurements. Each channel carries unit white noise. A narrow-band signal at 1.2 Hz, white noise through a second-order resonator with 2 % damping, is added at a signal-to-noise ratio of 3 or 10, once to a single channel and once to all channels in a shape containing a nodal line, the shape measured in the vertical setup of campaign 1. The first case gives $N_{\mathrm{eff}} \le 1.00$ without exception. The second gives ≥ 1.53 for five channels and ≥ 1.74 for six, in 95 % of trials.

A threshold of $N_{\mathrm{eff}} = 1.25$ separates them with no overlap in 400 trials per case. With three channels the two cases are not separated, and the threshold is not applied. Two departures from these idealised records were tested. Channel gains differing by up to $\pm 3\,\%$, the spread of the calibration factors, change N_{eff} by at most $4\,\%$. Noise levels differing between channels by up to a factor of ten leave the single-channel case below the threshold in every trial, but push $21\,\%$ of the shared-signal trials below it as well. The test can therefore reject a global mode recorded with unequal channel noise; it did not, in any trial, accept a local one. The quantity is the inverse participation ratio, familiar from studies of vibrational localisation in disordered media [82, 83]. No precedent was found for its use as an acceptance criterion in operational modal analysis, so it is defined explicitly here and justified by the simulations above rather than by citation. It is not a mass participation factor and should not be interpreted as one.

The second requirement is that a mode is the same for every observer. The eigenvalue depends on the mass, damping and stiffness of the structure, not on which degrees of freedom are instrumented; moving the sensors changes which components of the shape are observed and cannot change the eigenvalue. The criterion requires the same pole in every setup of a campaign and, where the measured degrees of freedom coincide, agreement of the shapes by the modal assurance criterion. A group of closely spaced modes, which Section 4 predicts for this foundation, cannot be matched pole by pole across setups that weight its members differently. The requirement is then applied to the group: every setup of a campaign must return a pole within it, and the group must be no wider than the model allows, about $20\,\%$ for the lowest lateral pair and torsion together, plus the scatter of a single estimate.

The third requirement is that a mode dissipates at a rate materials permit. Free decay follows an envelope $e^{-\xi\omega t}$, so a mode falls to $1/e$ of its amplitude in $1/(2\pi\xi)$ cycles. At $\xi = 0.01\,\%$ that is about 1600 cycles; at $\xi = 0.007\,\%$ and 100 Hz it is about 2300 cycles, or 23 seconds of sustained oscillation. A built structure under ambient excitation does not sustain free vibration for thousands of cycles, whereas a deterministic harmonic of constant frequency and amplitude presents to a covariance-driven estimator as exactly such a pole, its apparent damping reflecting estimator resolution rather than dissipation [69, 73]. The screening bound used here, $\xi = 0.1\,\%$, is adopted on that basis. Published field damping estimates for offshore support structures vary with load level and operating state [28], so no field value is adopted. The bound coincides instead with the lowest damping that EN 1991-1-4 tabulates for any structural element, a logarithmic decrement of 0.006 for parallel cables, against 0.03 to 0.10 ($\xi \approx 0.5$ to $1.6\,\%$) for reinforced concrete structures [85]. It is used only to reject, and it lies two orders of magnitude below the values the admitted band returns.

5.2 Classification of the identified poles

Applied to these records, the procedure returns poles in three well-separated regions. There is a group between 1.05 and 1.33 Hz and a set between 14 and 24 Hz. A third set, between 57 and 700 Hz, appears only under a configuration that retains the acquisition rate. Table 4 lists the accepted poles of the first two by setup. Figure 7 shows the stabilisation diagram of one setup and illustrates the difficulty: both the 1.2 Hz pole and the 18–20 Hz cluster stabilise cleanly over the full range of model orders, so nothing in the stabilisation behaviour distinguishes them.

Under the first requirement, Figure 8 gives the effective channel count for both bands in all seven setups. In the 13–24 Hz band five of the seven fall at or below 1.17, four of them at exactly 1.00: one channel responds and the rest lie at noise level, so there is no spatial pattern to interpret. In the 1.0–1.4 Hz band six of the seven exceed the threshold, ranging from 1.55 to 5.00. The low-band values are not uniform, and uniformity is not to be expected, since a pattern containing a nodal line necessarily leaves channels near that line with small amplitude, which lowers N_{eff} without implying locality. Two exceptions are reported rather than excluded. Lateral setup 2 of campaign 1 gives 1.00 in the low band as well; it is also the record with the strongest impulsive content, its narrow-band kurtosis [71] lying between 128 and 164, and the

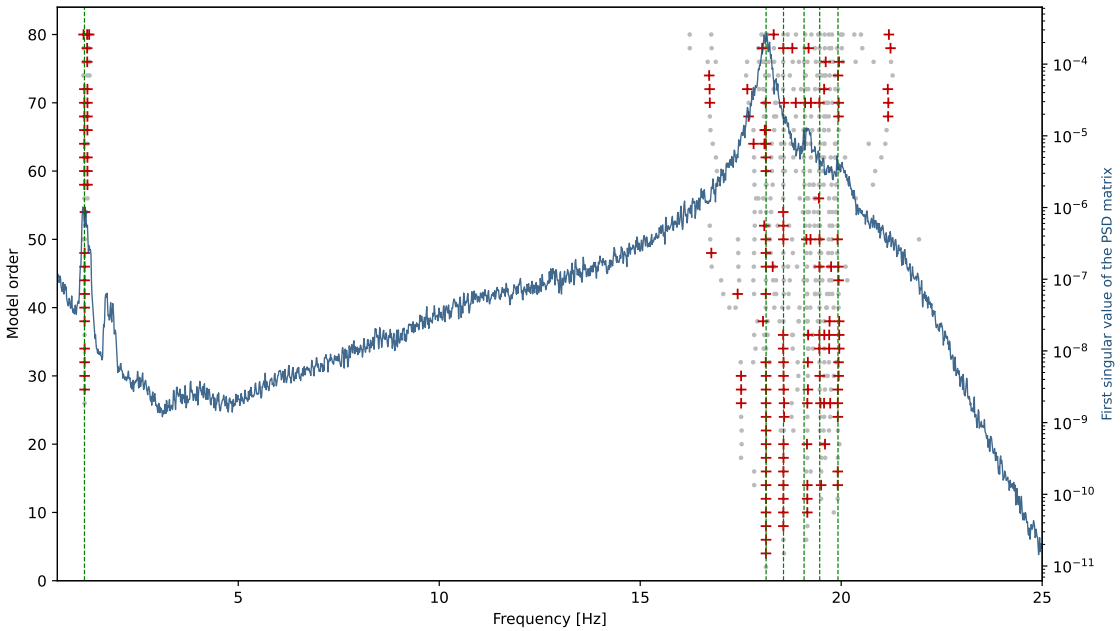


Figure 7: SSI-COV stabilisation diagram for the vertical setup of campaign 1, with the first singular value of the cross-power spectral density matrix overlaid. Both the 1.2 Hz pole and the 18–20 Hz cluster stabilise cleanly over the full range of model orders, so the stabilisation behaviour alone does not distinguish them

Table 4: Accepted poles by setup; frequency in Hz with damping ratio in per cent in parentheses.

Camp.	Setup	Low band	Higher band
1	Vertical	1.174 (4.74)	18.132 (0.70), 18.565 (0.68), 19.077 (0.58), 19.466 (0.59), 19.920 (1.30)
1	Lateral 1	1.226 (3.43), 1.254 (2.77), 1.731 (5.49)	21.920 (0.40), 22.751 (0.51), 23.754 (0.30)
1	Lateral 2	1.322 (2.71)	21.574 (5.95), 22.813 (1.25), 23.173 (0.59)
1	Three-channel	1.238 (3.01), 1.331 (4.39), 1.726 (5.35)	14.438 (1.42), 14.996 (0.97)
2	Lateral	1.052 (2.69), 1.215 (1.93), 1.612 (2.42)	none accepted
2	Vertical d1	1.147 (9.85)	11.806 (11.52), 13.871 (3.02), 16.833 (2.73), 17.087 (0.59), 17.673 (1.96), 18.954 (1.08), 19.473 (1.37), 19.689 (1.11), 20.549 (0.54)
2	Vertical d2ke	1.074 (1.25), 1.199 (2.35), 1.231 (2.48)	2.296 (1.19)

one whose channel RMS drifts by a factor of 112 (Table 2). That is the condition under which the simulations show the test rejecting a global mode, but the record cannot show that it did, so its low-band pole fails the first requirement and is not used in any comparison that follows. The three-channel setup gives 0.60 of its channel count in both bands, which with only three channels does not separate the two cases.

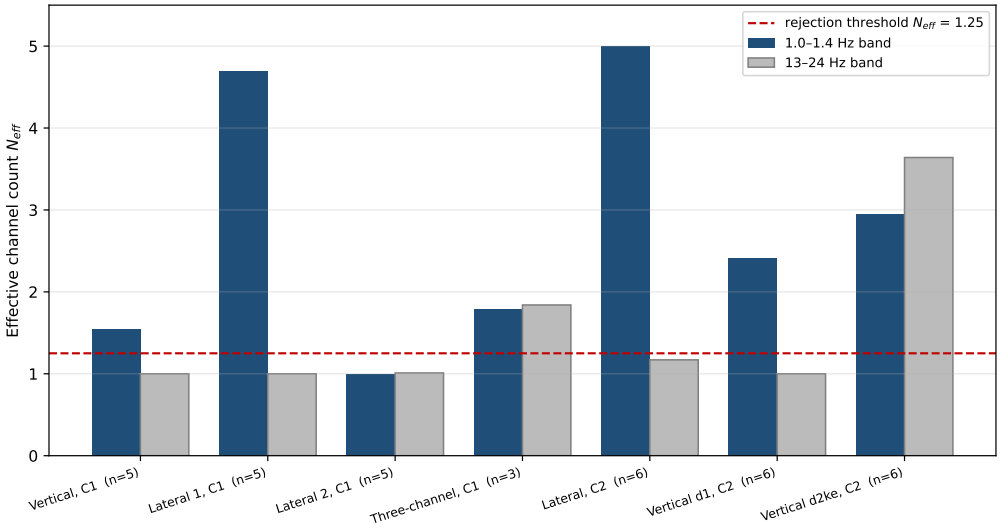


Figure 8: Effective channel count N_{eff} by setup and band, against the rejection threshold of 1.25 established by simulation in Section 5.1. The number of channels in each setup is given in the axis label. A value of unity means that one channel carries the entire response

Under the second requirement, the low-band poles within a campaign spread by ±6.3 % and ±7.7 %. That is an order of magnitude more than the scatter within a stabilisation cluster, so frequency agreement alone cannot settle the question. The criterion is therefore applied to the shapes. In all three vertical setups of both campaigns, opposite corners of the instrumented area move in antiphase, by 176°, 175° and 179°, about a nodal line through the centre of that area. In the lateral setups all horizontal channels move in phase and with similar amplitude. The two vertical setups of campaign 2 instrument five points in common, and between their corresponding clusters the modal assurance criterion is 0.967. The sixth channel of each sits at a different point and is left out of that comparison. Table 5 gives the phase relations and Figure 9 the patterns. One further observation supports this interpretation. The single vertically oriented channel in the lateral setup is also the only channel there that is out of phase and of small amplitude, and it was not selected for that purpose.

The scatter of a single estimate was measured by identifying each record again in three equal segments with the same settings. The low-band poles of one setup move between segments by 1 to 3 % from their mean, and by less than 0.3 % in the three-channel setup, the one record close to stationary. Lateral setup 2 of campaign 1 is the exception, with segment poles spread over 1.19 to 1.36 Hz. The low group spans 1.174 to 1.331 Hz in campaign 1 and 1.052 to 1.231 Hz in campaign 2, 13 and 16 % of its centre, and every setup of each campaign contributes to it; it satisfies the group form of the second requirement.

The 14–24 Hz poles do not. Within campaign 1, three setups recorded within ninety minutes return three disjoint bands, 18.1 to 19.9 Hz, 21.6 to 23.8 Hz and 14.4 to 15.0 Hz, spanning 65 % of their centre, and in campaign 2 two of the three setups return no pole above 5 Hz at all. These poles are local in the sense of the first requirement as well. Two sources are plausible. The first bending mode of an individual pile, over the fixity lengths inferred in Section 6.2 and carrying its added water, lies between about 8 Hz (one end pinned) and 17 Hz (both ends fixed), overlapping the lower part of the band. The response of the instrument mounting, which was

not documented, may lie anywhere in it [80]. The records cannot separate the two, and neither would be a mode of the cap. The verdict on this band is therefore that it is not a global mode of the foundation, not that its source is known.

Three further sets of poles fail the second requirement and are reported without interpretation. A pole at 1.61 to 1.73 Hz appears in three of the seven setups and has no counterpart in the model of Section 4, which places nothing between torsion and rocking. The extended search described below returns single poles at 0.69 Hz in lateral setup 1 of campaign 1 and at 0.83 Hz in the lateral setup of campaign 2, each in one setup only. And the vertical d2ke setup returns a pole at 2.30 Hz that no other setup shares.

One observation appears to tell against the shape agreement reported above. If instead of matching corresponding clusters one compares the first singular vector at the dominant low-band peak of each of the two vertical setups of campaign 2, the modal assurance criterion is 0.09, not 0.97. The peaks are at 1.210 and 1.229 Hz, and the two setups are therefore dominated by different members of the closely spaced group. The shapes of corresponding poles agree; the dominant pole differs between setups. Both facts are consequences of the group containing more than one mode, and Section 6.1 returns to them.

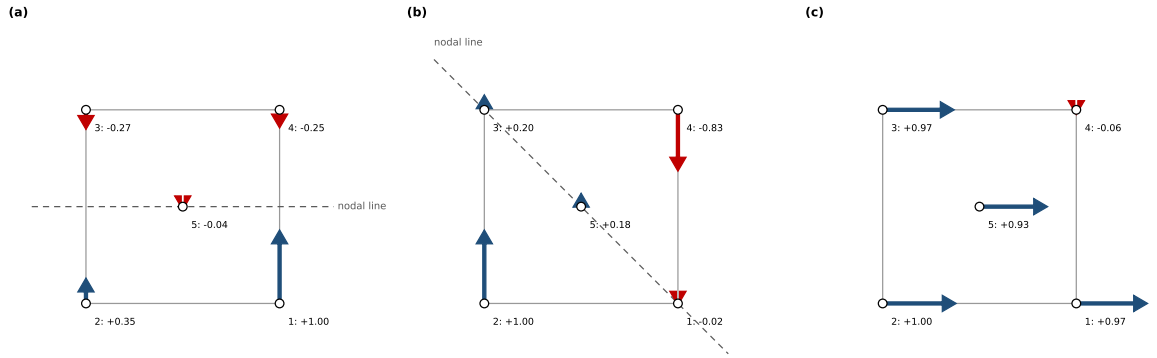


Figure 9: Measured deformation pattern of the 1.05–1.33 Hz band, taken from the first singular vector at the band peak. (a) Vertical setup of campaign 1. (b) Vertical setup d2ke of campaign 2. (c) Lateral setup of campaign 2. Signed amplitudes are given beside each node. The vertical setups show opposite corners in antiphase about a nodal line; the lateral setup shows the horizontal channels in phase, channel 4 being the only vertically oriented channel there

Table 5: Phase between opposite corners of the instrumented frame at the low-band peak, and amplitude at the centre relative to the mean of the corners. In the lateral setup, channel 4 is the only vertically oriented channel, which is why the 2–4 pair behaves unlike the 1–3 pair there.

Setup	Direction	Peak (Hz)	1 vs 3	2 vs 4	Centre / corners
Vertical, campaign 1	vertical	1.134	−179.1°	−173.3°	0.048 / 0.466
Vertical d1, campaign 2	vertical	1.210	+176.3°	+176.6°	0.424 / 0.517
Vertical d2ke, campaign 2	vertical	1.229	−142.7°	−175.2°	0.183 / 0.518
Lateral, campaign 2	horizontal	1.203	−4.9°	−143.3°	0.927 / 0.757

The third requirement needs the influence of the processing configuration settled first. Two processing choices govern what the identification can return, the covariance window and the high-pass cut-off, and varying them together would not separate their effects. One record was therefore processed under all four combinations, A to D of Table 6. Configuration C removes the filter while keeping the short window and still returns nothing below 5 Hz; configuration D lengthens the window while keeping the filter and likewise returns nothing. Only their joint

removal recovers the band. In this factorial the window changes together with the sampling rate, and configurations A and C also contain the mains harmonics, which take up model order. Configuration E therefore keeps the sampling rate and the filter of B and shortens only the window, to $i = 5$. It returns nothing below 5 Hz, but also no stable pole anywhere, since five block rows leave little room for any; it is consistent with the effect of the window without isolating it cleanly. The comparison supports the window bound of Section 5.1 as a constraint separate from the filter, but it does not establish the bound as a sharp limit.

Table 6: Comparison of processing configurations, one record (lateral setup, campaign 2). Configurations A–D form a factorial in window and high-pass cut-off, in which the window changes together with the sampling rate; E differs from B in the window alone.

Config.	Sampling rate	High-pass	i	τ	Poles returned
A	1651.6 Hz	5 Hz	150	0.091 s	20, from 57.6 to 700.9 Hz; none below 5 Hz
C	1651.6 Hz	0.3 Hz	150	0.091 s	19, from 80.9 to 604.6 Hz; none below 5 Hz
D	51.6 Hz	5 Hz	120	2.33 s	6, from 6.8 to 20.6 Hz; none below 5 Hz
B	51.6 Hz	0.3 Hz	120	2.33 s	3: 1.052, 1.215, 1.612 Hz
E	51.6 Hz	0.3 Hz	5	0.097 s	none

Seven of the twenty poles returned by configuration A carry damping between 0.007 % and 0.042 %, which the third requirement alone rejects. Only afterwards is the spectrum consulted, and it identifies them: the same seven lie within 0.2 % of an exact multiple of 50 Hz, at 100.166, 150.295, 200.284, 250.264, 300.488, 400.686 and 700.926 Hz, and six of them survive into configuration C. Divided by their harmonic orders they give a fundamental between 50.05 and 50.10 Hz, within the normal excursion of a 50 Hz supply. They are harmonics of the mains supply, which these records carry at high level, as Figure 10 shows.

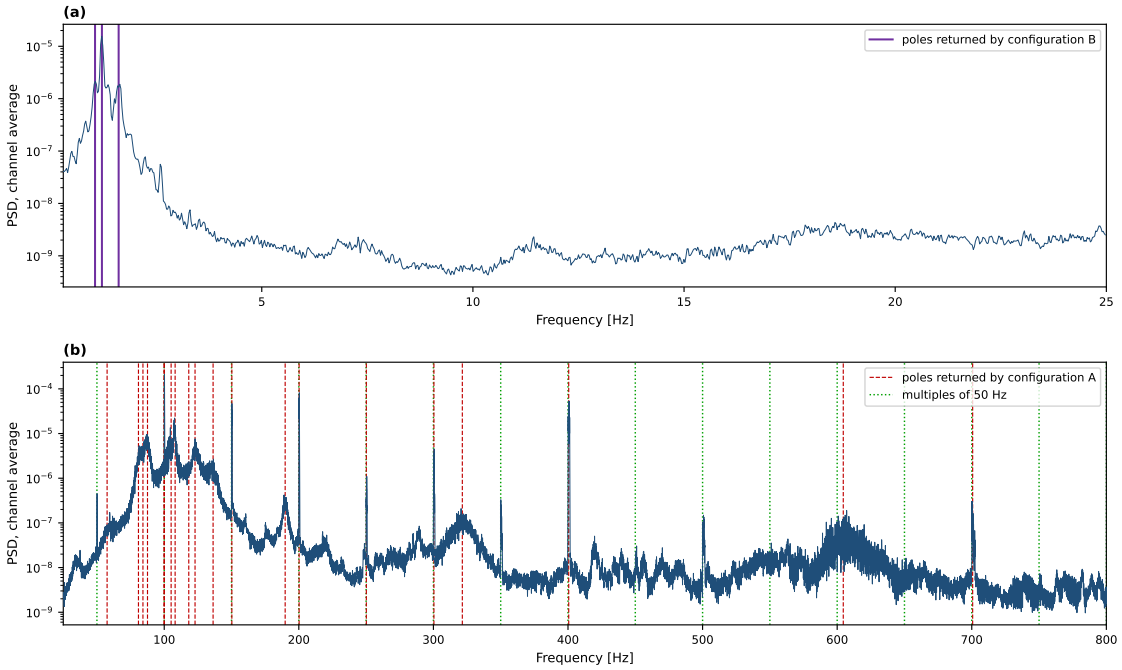


Figure 10: The same record seen by the two processing configurations of Table 6, over (a) the structural band from 0.3 to 25 Hz and (b) the band from 25 to 800 Hz from which configuration A draws its poles. Dotted lines mark multiples of 50 Hz

The region below the analysis band remains to be examined. The band searched begins at

0.5 Hz, which would exclude a tower mode: soft-stiff design places the fundamental mode of a tower of this class between the $1P$ and $3P$ rotor harmonics, near 0.3 Hz [23]. The records were therefore reanalysed with a further decimation to 12.9 Hz and $i = 150$, a window of 11.6 s or 3.5 periods at 0.3 Hz, and searched from 0.18 Hz. No pole below 0.5 Hz satisfies the criteria in any setup of either campaign. The only features there are spectral peaks near 0.21 Hz in campaign 1 and 0.32 Hz in campaign 2, each consistent between the setups of its own campaign; neither yields a stable pole. The 0.21 Hz peak was recorded before the tower stood and cannot belong to it; with a period of 4.7 s it is consistent with wave excitation. The 0.32 Hz peak lies where the first mode of the tower is expected, and it may be that mode seen weakly at the cap. It is not interpreted further, because the cap records alone cannot separate a tower mode from wave excitation at that frequency. Figure 11 shows the search.

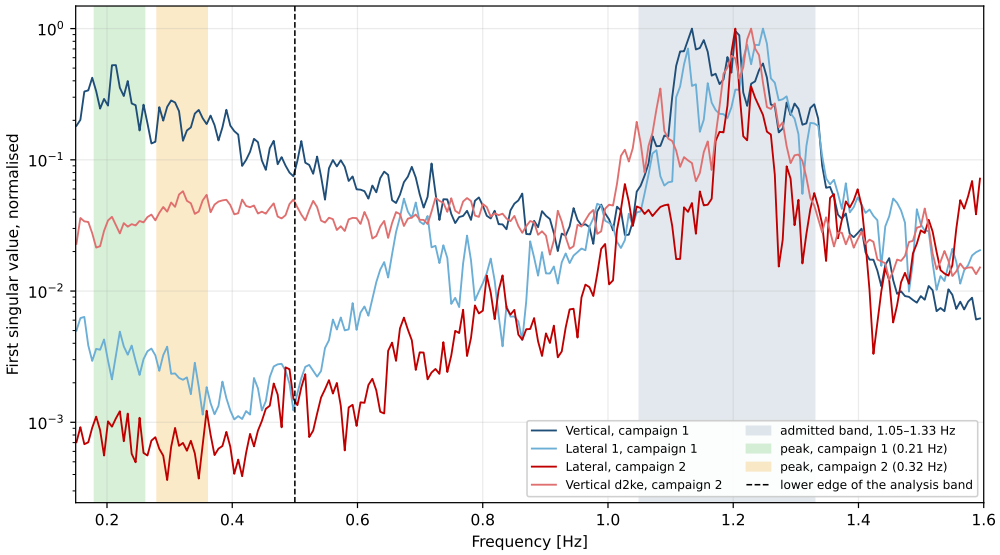


Figure 11: Search below the analysis band, with the records decimated to 12.9 Hz and a covariance window of 11.6 s. The only features below 0.5 Hz are the peaks marked at 0.21 Hz in campaign 1 and 0.32 Hz in campaign 2. The first, recorded before the tower stood, is consistent with wave excitation; the second lies where the first tower mode is expected. Neither yields a stable pole

Table 7 summarises the three requirements applied together. One band survives all three requirements; the other two fail on grounds independent of each other and of the identification settings.

Table 7: Classification of the three groups of poles.

Group	N_{eff}/n	Same in every setup	Damping	Verdict
1.05–1.33 Hz	0.31–0.94 in six of seven	yes, as a group; MAC 0.967 between corresponding clusters	1.3–9.9 %	structural
14–24 Hz	$1/n$ (one channel) in five of seven	no; three disjoint bands in one campaign, none in two setups of the other	0.3–11.5 %	local; not a mode of the cap
57–700 Hz	—	—	0.007–0.042 %	mains harmonics

6 Physical interpretation of the identified band

Sections 4 and 5 are independent of one another: the first uses only the geometry of the foundation, the second only the recorded motion. Combining them tests whether the band admitted by the records has the shape the geometry predicts. It also yields a quantity that neither line of evidence gives alone.

6.1 The lowest mode as rotation about the focal point

Section 4 predicts that the lowest band of this foundation is not a single mode. It should contain the lowest lateral pair, rotating about centres 48 to 105 m above the pile heads, and a torsional mode close to it. Rocking should lie five to sixteen times higher. The admitted band, 1.05 to 1.33 Hz, is the only low-frequency band in the records, and its spread within a campaign already exceeds the identification scatter by an order of magnitude. Three tests compare it with the prediction.

The vertical setups do not discriminate. In every vertical setup opposite corners of the frame move in antiphase about a nodal line (Section 5.2). That is the vertical component of a rotation about a horizontal axis. Focal-point sway and rocking both produce it, and so does any rotation about a centre away from the measured surface; only the height of the centre separates them. The vertical pattern establishes that the band involves rotation about a horizontal axis and excludes pure torsion as its dominant member, but no more.

The lateral setup of campaign 2 measures the height of the centre. Its processing log records the orientation of every channel: channels 1, 2, 3, 5 and 6 measure horizontally in one direction and channel 4 vertically. The ratio of the vertical amplitude at channel 4 to the mean horizontal amplitude is therefore the quantity bounded by (4). Table 8 gives it from three estimators. The first singular vector at the band peak gives 0.084. The ratio of power spectral densities gives 0.116, but the coherence between channel 4 and the central horizontal channel is only 0.50 there, so part of that amplitude is noise; weighting by the coherence gives 0.082. The SSI shape of the translational pole at 1.215 Hz gives 0.16. Focal-point sway predicts 0.025 to 0.081 for a channel 2.5 to 3.5 m from the rotation axis, and rocking 0.8 to 1.5. The measurement lies at the upper end of the first range and an order of magnitude below the second. A residual vertical reading of a few per cent is also expected from the transverse sensitivity of the accelerometer and from any offset of the frame from the cap centre, which would increase d.

Rocking could reproduce a ratio as small as 0.16 only if the cap moved within about ±6° of the one direction that places channel 4 on the rotation axis. The orientation of the frame relative to the motion is not known, so this cannot be excluded outright. It would, however, require a coincidence that the vertical setups do not suggest, since their nodal lines take different orientations. The two lateral setups of campaign 1 cannot be used for this test, because the orientation of their channel 4 is not documented.

The band contains a torsional member. The same lateral setup returns two poles in the band, and their shapes differ in kind. At 1.215 Hz the four horizontal channels move in phase, within 8°, with amplitudes between 0.96 and 1.15 of the central channel: a translation. At 1.052 Hz, channels 1 and 2, on one edge of the frame, move at 165° to channel 3 on the opposite edge and to the central channel. Horizontal sensors sharing one direction see a rotation about a vertical axis in exactly this way, with opposite edges in antiphase. The pole therefore has a strong torsional component. This is the composition Section 4 predicts: torsion close to the lowest lateral pair, because both are resisted mainly by pile bending. The model places torsion 9 to 21 % above the lateral pair, whereas the measured torsional member lies 13 % below the translational one. The model fixes the proximity of the two but not their order, which depends on how the soil restrains a pile radially and tangentially and, in campaign 2, on the tower above the cap.

Table 8: Vertical-to-horizontal amplitude ratio in the lateral setup of campaign 2, against the prediction of Section 4 for a channel 2.5 to 3.5 m from the rotation axis. The horizontal amplitude is the mean of channels 1, 2, 3 and 5

Estimator	f (Hz)	Vertical / horizontal
First singular vector at the band peak	1.203	0.084
Power spectral density ratio at the peak	1.203	0.116
coherence of channels 4 and 5; coherence-weighted ratio		0.50 ; 0.082
SSI shape, translational pole	1.215	0.160
SSI shape, torsional pole	1.052	0.139
Predicted, lowest lateral pair		0.025–0.081
Predicted, rocking pair		0.8–1.5

The predicted rocking and vertical modes were not identified. With the axial stiffness of Section 6.2 they fall between about 7.5 and 12 Hz. Configuration D of Table 6 returns poles from 6.8 to 20.6 Hz, and the vertical d1 setup an 11.8 Hz pole, but none of them recurs in a second setup, so none satisfies the second requirement. Ambient excitation at the cap evidently excites them weakly, and the records do not establish them.

Two qualifications apply. The position of the frame on the cap is not documented, and the distance d in (4) rests on the inference of Section 3.2 that the frame is concentric. The discriminating test, moreover, rests on one setup, the only one whose channel orientations are recorded. Taken together, the three tests support the prediction: a near-translational lowest pair rotating about an elevated centre, with torsion close to it. They exclude rocking unless the motion happened to be aligned with the one direction that hides it.

6.2 Depth to fixity of the piles

Section 4.2 showed that the lowest lateral pair is governed by pile bending and hardly at all by the axial pile stiffness. The measured frequency therefore cannot be inverted for k_{ax}, as an uncoupled sway model would suggest. It can be inverted for the effective fixity length, once k_{ax} is bounded independently.

The axial stiffness is estimated forward from the borehole. The shear modulus of each

layer follows from its SPT value through the correlation of Ohsaki and Iwasaki [86], $G_0 = 1200\,N^{0.8}$ t/m^2, or about $11.8\,N^{0.8}$ MPa. The correlation was derived from seismic exploration, so it gives the small-strain modulus, which is the modulus that ambient vibration mobilises. A lower bound of $0.5\,G_0$ allows for the scatter of the correlation and for disturbance of the soft clay by pile installation. The solution of Randolph and Wroth [4] for a compressible pile in non-homogeneous soil then gives the stiffness of the embedded length; placed in series with the free length, it gives 4.7 to 5.8×10^8 N/m for an isolated pile. Pile–soil–pile interaction acts on the embedded length only. With the axial interaction factors of Randolph and Wroth [5] assembled over the 34 pile positions, the head flexibility becomes $\mathbf{F} = \mathbf{I}/k_{\text{free}} + \mathbf{A}/k_{\text{emb}}$. That reduces the effective stiffness per pile to 2.5 to 3.4×10^8 N/m. The model of Section 4 is evaluated over 2.5 to 5.8×10^8 N/m, a range over which the lowest lateral pair changes by about 3 %.

Inverting the setup means of campaign 1, 1.17 to 1.32 Hz, over that range of k_{ax} and over the bounds of the participating pile mass gives L_{eff} between 13.9 and 17.0 m. Since 12.34 m of that is free length in water, the piles are effectively fixed 1.6 to 4.7 diameters below the seabed. At the campaign mean of 1.255 Hz the range narrows to 2.1 to 3.8 diameters.

This can be checked against an estimate that uses no measured frequency. For a pile in soil of constant modulus, Davisson and Robinson [87] place the point of equivalent fixity at $1.4R$ below the ground surface, with $R = (EI/E_s)^{1/4}$. Taking $E_s = 3G$, appropriate to undrained clay, with G averaged over the upper 3 to 5 m of the profile at 0.5 to 1.0 times G_0, gives 2.3 to 3.0 diameters. The two ranges overlap. The forward estimate rests on an SPT correlation applied at $N \leq 2$, where it is least reliable, and on a rule derived for static loading. Agreement at this level is as much as the information supports, and the comparison is an order-of-magnitude check. Table 9 collects the values.

Table 9: Axial pile stiffness used as input, and the depth to fixity inverted from the measured frequency and estimated forward from the soil profile.

Axial stiffness, isolated pile, $G = 0.5$–$1.0\,G_0$	4.7–5.8×10^8 N/m
Axial stiffness, in the group, interaction on the embedded length	2.5–3.4×10^8 N/m
Change of f_1 over 2.5–5.8×10^8 N/m	about 3 %
Depth to fixity, inverted from $f = 1.17$–1.32 Hz	1.6–4.7 pile diameters
at the campaign 1 mean, 1.255 Hz	2.1–3.8 pile diameters
Depth to fixity, Davisson and Robinson, $E_s = 3G$, $G = 0.5$–$1.0\,G_0$	2.3–3.0 pile diameters

6.3 Effect of tower installation

Campaign 2 was recorded after the tower was erected. The model of Section 4 is a model of the bare cap. The tower adds mass and rotational inertia far above the cap and its own flexibility, so the model does not predict the campaign 2 frequencies. The question here is narrower: whether the records show a change in the band between the two campaigns.

Setup means cannot answer it directly, because the setups of the two campaigns differ in direction, and the band contains members of different kinds that different directions weight differently. The means of all setups fall by 8.4 % between campaigns, from 1.255 to 1.150 Hz, but that comparison mixes three vertical setups with two lateral ones and a mixed one. Table 10 and Figure 12 therefore group the setups by measured direction.

Like-for-like comparisons give a much smaller change. The vertical setup of campaign 1 and the two vertical setups of campaign 2 differ by -2.3 and -0.5 %. In the lateral setups, the translational member of campaign 2, at 1.215 Hz, lies 2.0 % below the mean of lateral setup

1 of campaign 1. The lateral comparison rises to −8.6 % only when the torsional member at 1.052 Hz is averaged in. It rises to −14 % against lateral setup 2, which fails the first requirement. Changes of 0.5 to 2.3 % are no larger than the 1 to 3 % by which a single pole moves between segments of one record (Section 5.2), and well within the spread of the band within one campaign, ±6 to ±8 %. The records therefore show no change attributable to the tower that exceeds the resolution of these campaigns. No participating mass is inferred.

Three further limitations apply. The records do not document the state of construction at campaign 2 beyond the presence of the tower. No environmental or operational data, tide level among them, accompany either campaign, and the free length of the piles in water changes with the tide. And the 0.32 Hz peak of campaign 2, which may be the first tower mode (Section 5.2), shows that the tower does contribute to the cap response at its own frequency.

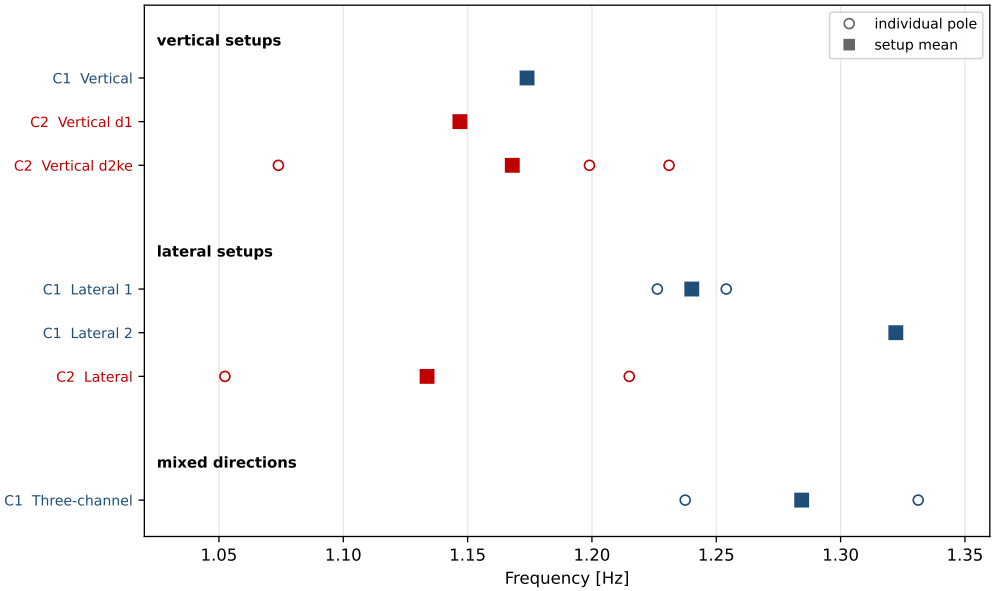


Figure 12: Poles of the 1.05–1.33 Hz band by setup, grouped by measured direction, with setup means. Blue: campaign 1, before the tower; red: campaign 2, after it. Like-for-like setups differ by 0.5 to 2.3 %. The larger difference between the lateral setups comes from the torsional member at 1.052 Hz in campaign 2

Table 10: Low-band poles by setup, grouped by measured direction, and the change between campaigns for like-for-like setups. V: all channels vertical; H: horizontal, with channel 4 vertical in campaign 2 and undocumented in campaign 1; M: one vertical and two horizontal channels.

Dir.	Camp.	Setup	Poles (Hz)	Setup mean (Hz)
V	1	Vertical	1.174	1.174
V	2	Vertical d1	1.147	1.147
V	2	Vertical d2ke	1.074, 1.199, 1.231	1.168
		change, vertical: −2.3 % and −0.5 %		
H	1	Lateral 1	1.226, 1.254	1.240
H	1	Lateral 2	1.322	1.322
H	2	Lateral	1.052 (torsional), 1.215 (translational)	1.134
		change, translational member against lateral 1: −2.0 %		
M	1	Three-channel	1.238, 1.331	1.284

7 Conclusions

The lowest mode of an elevated cap on inclined piles has a shape that the geometry largely fixes. The axial pile springs, which carry most of the soil stiffness into the problem, are nearly concurrent: the axes of each ring meet at a point above the cap. Rotation about that point stretches no pile. The lowest lateral mode therefore rotates about a centre at or above the focal point and is resisted mainly by pile bending. At the cap it is nearly a translation, with a vertical component that the setting-out drawing alone bounds. Rocking lies five to sixteen times higher, and torsion, resisted by bending alone, lies close to the lowest lateral pair. These statements hold whatever the axial pile stiffness. The frequencies do not: they depend on the soil through the depth to fixity, and the measured frequency constrains that depth rather than the axial stiffness.

The records of the T18 foundation agree with this picture as far as they can test it. One band, 1.05 to 1.33 Hz, satisfies the three requirements placed on a physical mode. In the one setup whose channel orientations are documented, its vertical component is 0.08 to 0.16 of its horizontal motion, against 0.03 to 0.08 predicted for rotation about the focal point and about 1 for rocking. One member of the band moves the cap in torsion, as the model expects. The comparison rests on a single setup and on an undocumented frame position, and it leaves the order of the torsional and lateral members unresolved.

The same records show how readily a measurement of this kind may be misinterpreted. The most prominent poles in them are not modes of the foundation. Some move a single sensor and shift when the sensors are moved; others decay too slowly to belong to a structure and prove to be interference from the mains supply. Which of them an identification returns depends as much on the settings the analyst chooses as on the foundation itself. Acceptance was therefore made to rest on the properties a mode must possess rather than on how cleanly a pole stabilises. A frequency change between campaigns, likewise, is meaningful only between setups that measure the same direction. Compared that way, the change after tower installation does not exceed the resolution of these campaigns.

Statements and Declarations

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Author Contributions Nguyen Hoang Bui: conceptualization, methodology, software, formal analysis, investigation, visualization, writing – original draft. Thi Cam Nhung Nguyen: conceptualization, methodology, investigation, resources, data curation, project administration, supervision, writing – review and editing. Huu-Quyet Nguyen: investigation, data curation. Chung Truong Thanh: software, validation. Hung Nguyen-Dang: writing – review and editing. All authors read and approved the final manuscript.

Data Availability Statement The acceleration records analysed in this study are available from the corresponding author upon reasonable request by e-mail. The design drawings belong to the project owner and are not redistributed; every quantity read from them is given in Table 1 and Figure 2.

Code Availability The analysis is implemented in Python (NumPy, SciPy, Matplotlib). The code that regenerates every reported quantity, table and figure from the records is publicly available at `https://github.com/chungntu/t18-foundation-oma`.

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